

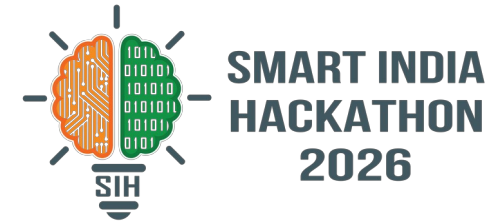
NISHAN-PQ

At-decryption session provenance for air-gapped documents.

PROBLEM STATEMENT ID	SIH26237
PROBLEM STATEMENT TITLE	Cryptographic Attribution and Immutable Decryption Provenance for Multi-Recipient Encrypted Document Distribution
THEME	Blockchain & Cybersecurity
PS CATEGORY	Software
TEAM ID	[ENTER TEAM ID]
TEAM NAME (REGISTERED ON PORTAL)	[ENTER REGISTERED TEAM NAME]



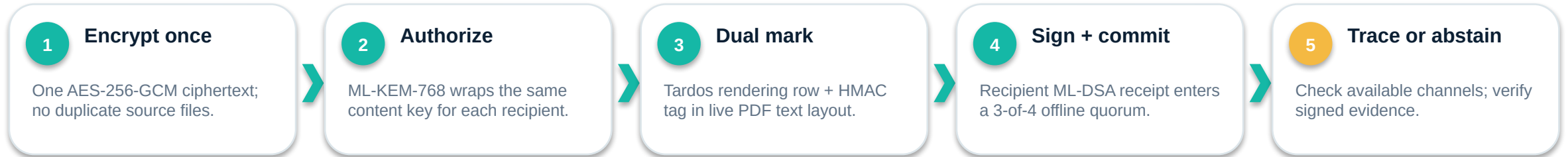
IDEA TITLE / PROPOSED SOLUTION



[TEAM NAME]

NISHAN-PQ Session-bound evidence with explicit abstention

Bind visual evidence + optional layout evidence + a recipient signature + a co-located quorum receipt to each released session.



DIGITAL DEVELOPMENT

30/30 exact JPEG-Q55 sessions

- All 1,000 rows scored; 0 extra rows accused
- Mean 13,433.337; population SD 128.216
- 30 codebooks × 4 strategies = 120 code-level cases; not physical trials

REAL CAPTURES

Historical 0/4 · Experimental 1/4 per profile

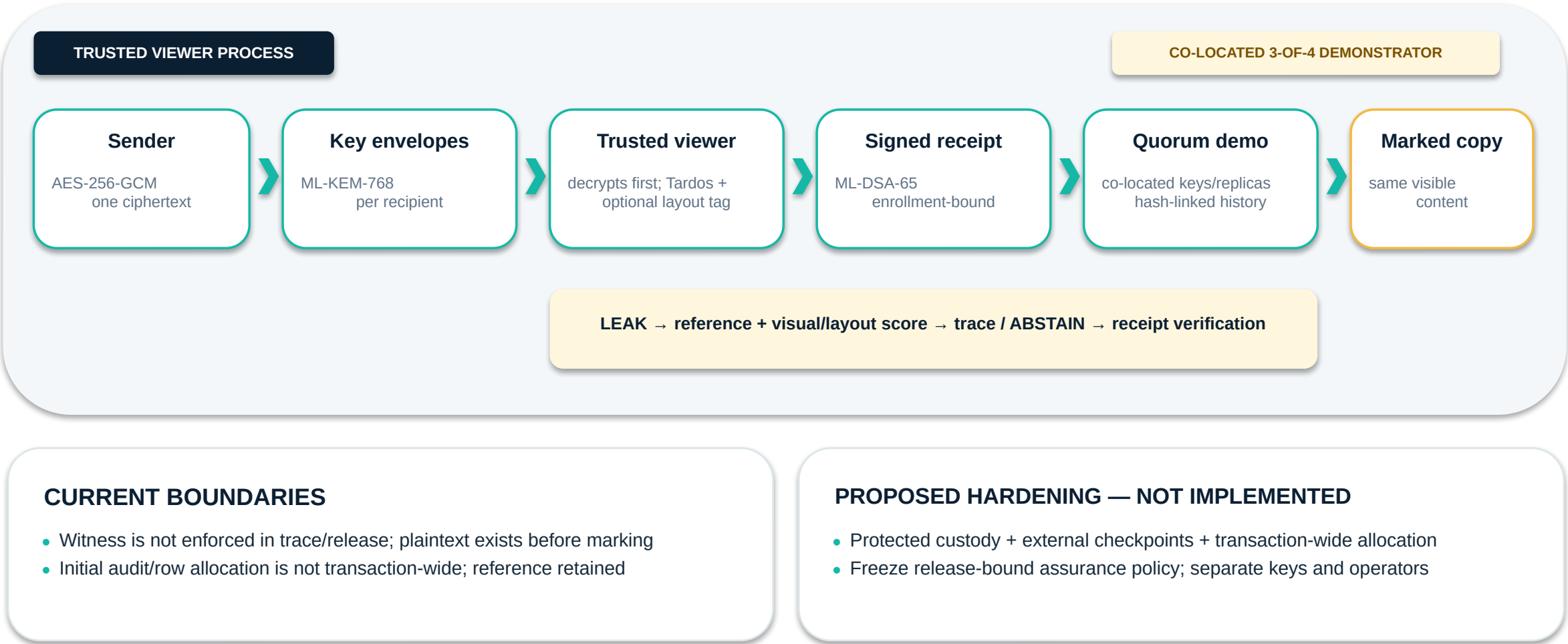
- Latest akshay3: 1,827.334 vs 1,096.290
- Same akshay3 capture recovered; refinement adds no capture
- Four captures · one printed single-page public fixture
- Visual code-row research; not signed end-to-end physical attribution

LIMIT: small, non-independent fixtures; these results are not a deployed reliability or population-safety guarantee.

TECHNICAL APPROACH

[TEAM NAME]

Implemented path, trusted-process assumption and demonstrator boundary



FEASIBILITY AND VIABILITY

[TEAM NAME]

COMPACT MEASURED OUTCOMES

- Digital: 30/30 exact JPEG-Q55 sessions; all 1,000 rows scored
- Physical: historical 0/4; conditional 1/4 per profile, same capture
- Searchable text retained; PDF bytes are not identical and signatures are not proven preserved
- Conflict/removal fixtures abstain; witness detects the recorded rollback fixture

41.87

MEAN PSNR · 5
PDFs

25

RECORDED DIGITAL
CASES

120/120

CODE TRIALS ·
ALL 5

EXPERIMENTAL EVIDENCE · NOT A RELIABILITY RATE

120 code-level cases are not 120 physical trials.

FOUR MATERIAL RISK GROUPS

HIGH

Physical

One page / four captures

UNRESOLVED · held-out print/camera matrix

HIGH

Raster / capacity

Low-capacity bypass; raster policy open

UNRESOLVED · fail/restrict release policy

HIGH

Rollback / row race

Witness separate; allocation not atomic

UNRESOLVED · external checkpoint + atomic row

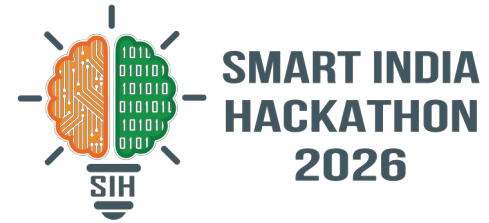
HIGH

Framing / removal

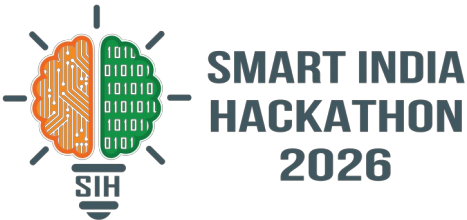
Authority can re-mark; both carriers removable

UNRESOLVED · custody + negative attacks

Recipient session evidence is not proof of who leaked it.



IMPACT AND BENEFITS



[TEAM NAME]

INVESTIGATIVE ASSISTANCE

Offline session leads with PQ-signed evidence

CURRENT POSITION

Research proposal — not ready for operational accusations

01 Session-level lead

Narrows an investigation to signed release sessions; it does not identify the human actor.

02 Offline operation

The measured demo does not depend on cloud APIs, tokens or a public chain.

03 PQC mechanism

ML-KEM and ML-DSA protect key delivery and release receipts in the prototype.

04 Evidence discipline

Conflict, removal and proof boundaries are recorded instead of hidden.

THREE GATES BEFORE OPERATIONAL USE

GATE 1

Security boundaries + defensible claim language

REQUIRES: Release policy; accusation limits

GATE 2

Independent documents, keys and recipients

REQUIRES: Multi-page + signed/tagged PDFs + negative attacks

GATE 3

Held-out physical comparisons

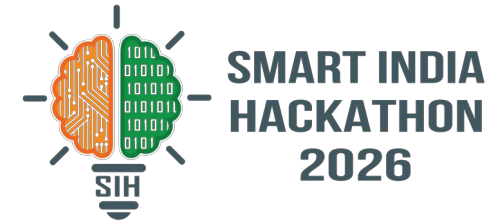
REQUIRES: Matched visibility + security budgets

RESEARCH AND REFERENCES

[TEAM NAME]

PRIMARY SOURCES & FOUNDATIONAL WORK

- 1 Official SIH26237 problem statement**
sih.gov.in/sih2026PS
- 2 NIST FIPS 203 / 204 — ML-KEM / ML-DSA**
csrc.nist.gov/pubs/fips/203/final · /fips/204/final
- 3 Tardos + T-Tracer + text-layout / dual marks**
ACM / IEEE sources; full URLs in notes
- 4 Physical/screen-camera baselines**
StegaStamp · arXiv:2304.12682
- 5 TRACE scope boundary**
arXiv:2607.08400 concerns LLM-agent trajectories
- 6 Saved physical + digital evidence**
research/evidence/.../scores.json · artifacts/nishan/*.json
- 7 Public repository**
[INSERT PUBLIC REPOSITORY LINK BEFORE UPLOAD]



PRIOR-ART BOUNDARY

KNOWN / PRESCRIBED

SIH26237 prescribes decryption marking, PQ signatures and DLT. Tardos, layout/dual marks and physical baselines pre-exist.



HONEST CONTRIBUTION

Measured integration plus failure analysis: session-bound dual-domain evidence, explicit abstention, and documented physical limitations—not exclusive novelty or best-in-world performance.

Shorter code / stronger carrier: unvalidated hypotheses